

Light

1. Light travels in _____.

- a) straight line
- b) curved path
- c) zigzag
- d) random

Correct answer: a) straight line

2. Reflection of light obeys _____.

- a) angle i = angle r
- b) $i > r$
- c) $i < r$
- d) $i + r = 90^\circ$

Correct answer: a) angle i = angle r

3. Laws of reflection are valid for _____.

- a) plane surfaces only
- b) curved only
- c) all reflecting surfaces
- d) mirrors only

Correct answer: c) all reflecting surfaces

4. Image in plane mirror is _____.

- a) real
- b) virtual
- c) inverted
- d) magnified

Correct answer: b) virtual

5. Plane mirror forms image as far _____.

- a) behind as object in front
- b) in front
- c) at focus

d) at infinity

Correct answer: a) behind as object in front

6. Minimum mirror length for full view _____.

a) half height

b) full height

c) quarter height

d) double height

Correct answer: a) half height

7. Concave mirror: reflecting surface _____.

a) curved inwards

b) outwards

c) plane

d) rough

Correct answer: a) curved inwards

8. Principal axis is line joining _____.

a) pole and centre of curvature

b) focus and pole

c) both

d) none

Correct answer: c) both

9. Pole is _____ of mirror.

a) centre

b) midpoint

c) edge

d) focus

Correct answer: b) midpoint

10. Centre of curvature for spherical mirror _____.

a) centre of sphere

b) focus

c) pole

d) radius

Correct answer: a) centre of sphere

11. Focal length $f = \underline{\hspace{2cm}}$ R.

a) $R/2$

b) $2R$

c) R

d) $R/4$

Correct answer: a) $R/2$

12. Sign convention: object distance u is $\underline{\hspace{2cm}}$.

a) always negative

b) positive

c) zero

d) depends

Correct answer: a) always negative

13. Real image is $\underline{\hspace{2cm}}$.

a) inverted

b) erect

c) virtual

d) enlarged

Correct answer: a) inverted

14. Virtual image is $\underline{\hspace{2cm}}$.

a) erect

b) inverted

c) real

d) cannot project

Correct answer: a) erect

15. For concave mirror, object at infinity: image at $\underline{\hspace{2cm}}$.

a) F

b) C

c) infinity

d) pole

Correct answer: a) F

16. Object beyond C: image _____.

a) between F and C, real, inverted, diminished

b) beyond C

c) behind mirror

d) at pole

Correct answer: a) between F and C, real, inverted, diminished

17. Object at C: image _____.

a) at C, real, inverted, same size

b) at F

c) infinity

d) behind

Correct answer: a) at C, real, inverted, same size

18. Object between C and F: image _____.

a) beyond C, real, inverted, enlarged

b) at F

c) behind

d) diminished

Correct answer: a) beyond C, real, inverted, enlarged

19. Object between F and P: image _____.

a) behind mirror, virtual, erect, enlarged

b) real

c) at C

d) diminished

Correct answer: a) behind mirror, virtual, erect, enlarged

20. Convex mirror always forms _____.

a) virtual, erect, diminished image

b) real

c) inverted

d) enlarged

Correct answer: a) virtual, erect, diminished image

21. Mirror formula: _____.

a) $1/f = 1/v + 1/u$

b) $1/v - 1/u = 1/f$

c) $f = v + u$

d) $v/u = f$

Correct answer: b) $1/v - 1/u = 1/f$

22. Magnification $m =$ _____.

a) $h_i/h_o = -v/u$

b) v/u

c) u/v

d) f/u

Correct answer: a) $h_i/h_o = -v/u$

23. $m > 1$ means _____.

a) enlarged

b) diminished

c) same size

d) inverted

Correct answer: a) enlarged

24. Refraction occurs due to _____.

a) change in speed of light

b) reflection

c) absorption

d) scattering

Correct answer: a) change in speed of light

25. Light bends _____ normal when entering denser medium.

a) towards

b) away

c) parallel

d) perpendicular

Correct answer: a) towards

26. Refractive index $n =$ _____.

a) c/v

b) v/c

c) $\sin i / \sin r$

d) all

Correct answer: d) all

27. Snell's law: $n_1 \sin i =$ _____.

a) $n_2 \sin r$

b) $\sin r / \sin i$

c) n_2 / n_1

d) constant

Correct answer: a) $n_2 \sin r$

28. Absolute refractive index of diamond _____.

a) 2.42

b) 1.5

c) 1.33

d) 1

Correct answer: a) 2.42

29. Relative refractive index air-glass _____.

a) 1.5

b) 2.42

c) 1.33

d) 1

Correct answer: a) 1.5

30. Critical angle for water _____.

a) 48.6°

b) 41.8°

c) 90°

d) 0°

Correct answer: a) 48.6°

31. Total internal reflection occurs when _____.

a) $i > c$, light to rarer

b) $i < c$

c) $r > 90^\circ$

d) normal incidence

Correct answer: a) $i > c$, light to rarer

32. Mirage occurs due to _____.

a) total internal reflection

b) refraction

c) reflection

d) dispersion

Correct answer: a) total internal reflection

33. Twinkling of stars due to _____.

a) refraction in atmosphere

b) reflection

c) total internal reflection

d) scattering

Correct answer: a) refraction in atmosphere

34. Convex lens: middle part _____.

a) thicker

b) thinner

c) equal

d) curved

Correct answer: a) thicker

35. Concave lens: middle part _____.

a) thinner

b) thicker

c) equal

d) plane

Correct answer: a) thinner

36. Lens maker formula for convex lens: _____.

a) $1/f = (n-1)(1/R_1 - 1/R_2)$

b) negative

c) zero

d) infinite

Correct answer: a) $1/f = (n-1)(1/R_1 - 1/R_2)$

37. Power of lens $P =$ _____.

a) $1/f$ (f in metre)

b) f

c) n-1

d) R

Correct answer: a) $1/f$ (f in metre)

38. Unit of power _____.

a) dioptre

b) watt

c) joule

d) ampere

Correct answer: a) dioptre

39. For convex lens, object beyond 2F: image _____.

a) between F and 2F, real, inverted, diminished

b) beyond 2F

c) virtual

d) erect

Correct answer: a) between F and 2F, real, inverted, diminished

40. Object at 2F: image _____.

a) at 2F, real, inverted, same size

b) at F

c) infinity

d) virtual

Correct answer: a) at 2F, real, inverted, same size

41. Object between 2F and F: image _____.

a) beyond 2F, real, inverted, enlarged

b) between F and 2F

c) virtual

d) erect

Correct answer: a) beyond 2F, real, inverted, enlarged

42. Object between F and lens: image _____.

a) same side, virtual, erect, enlarged

b) real

c) inverted

d) diminished

Correct answer: a) same side, virtual, erect, enlarged

43. Concave lens always forms _____.

a) virtual, erect, diminished

b) real

c) inverted

d) enlarged

Correct answer: a) virtual, erect, diminished

44. Dispersion: white light splits into _____.

a) 7 colours

b) 3 colours

c) 2 colours

d) no split

Correct answer: a) 7 colours

45. VIBGYOR order: violet _____ wavelength.

a) shortest

b) longest

c) equal

d) medium

Correct answer: a) shortest

46. Rainbow formed by _____.

a) dispersion + total internal reflection

b) refraction only

c) reflection

d) scattering

Correct answer: a) dispersion + total internal reflection

47. Human eye: cornea provides _____.

a) $\frac{2}{3}$ power

b) $\frac{1}{3}$ power

c) equal

d) zero

Correct answer: a) $\frac{2}{3}$ power

48. Near point of normal eye _____.

a) 25 cm

b) infinity

c) 10 cm

d) 50 cm

Correct answer: a) 25 cm

49. Far point _____.

a) infinity

b) 25 cm

c) 10 cm

d) 50 cm

Correct answer: a) infinity

50. Myopia corrected by _____.

a) concave lens

b) convex

c) bifocal

d) cylindrical

Correct answer: a) concave lens

51. Hypermetropia by _____.

a) convex lens

b) concave

c) plane

d) none

Correct answer: a) convex lens

52. Presbyopia needs _____.

a) bifocal lens

b) single

c) concave

d) convex

Correct answer: a) bifocal lens

53. Simple microscope: _____.

a) convex lens

b) concave

c) plane

d) combination

Correct answer: a) convex lens

54. Compound microscope has _____.

a) objective + eyepiece

b) single lens

c) mirror

d) prism

Correct answer: a) objective + eyepiece

55. Astronomical telescope: objective _____.

a) large focal length convex

b) short

c) concave

d) plane

Correct answer: a) large focal length convex

56. Angle of incidence _____ angle of reflection.

a) equals

b) greater

c) smaller

d) perpendicular

Correct answer: a) equals

57. Incident ray, reflected ray, normal lie in _____.

a) same plane

b) different planes

c) perpendicular

d) parallel

Correct answer: a) same plane

58. For regular reflection surface must be _____.

a) smooth

b) rough

c) curved

d) plane only

Correct answer: a) smooth

59. Lateral inversion in plane mirror _____.

a) left-right interchange

b) up-down

c) no inversion

d) inverted

Correct answer: a) left-right interchange

60. Sphere radius R , mirror focal length _____.

a) $R/2$

b) $2R$

c) R

d) infinite

Correct answer: a) $R/2$

61. Ray parallel to principal axis passes through _____.

a) focus after reflection (concave)

b) away (convex)

c) pole

d) centre

Correct answer: a) focus after reflection (concave)

62. Ray through focus after reflection _____.

a) parallel to axis

b) through C

c) pole

d) normal

Correct answer: a) parallel to axis

63. Ray through C reflects _____.

a) along same path

b) parallel

c) focus

d) pole

Correct answer: a) along same path

64. Convex mirror field of view _____.

a) wider

b) narrower

c) same

d) zero

Correct answer: a) wider

65. Used as rear view mirrors _____.

a) convex

b) concave

c) plane

d) all

Correct answer: a) convex

66. Refraction: speed _____ in denser medium.

a) decreases

b) increases

c) same

d) zero

Correct answer: a) decreases

67. Apparent depth formula _____.

a) real depth / n

b) $n \times \text{depth}$

c) $\text{depth} \times n$

d) same

Correct answer: a) real depth / n

68. Pencil in water appears _____.

a) bent

b) straight

c) broken

d) erect

Correct answer: a) bent

69. Diamond sparkles due to _____.

a) high n, TIR

b) low n

c) reflection

d) absorption

Correct answer: a) high n, TIR

70. Optical fibre uses _____.

a) TIR

b) refraction

c) reflection

d) dispersion

Correct answer: a) TIR

71. Convex lens converges rays to _____.

a) focus

b) diverges

c) parallel

d) infinity

Correct answer: a) focus

72. Power +20 D means _____.

a) convex lens $f=5$ cm

b) concave

c) plane

d) diverging

Correct answer: a) convex lens $f=5$ cm

73. Two thin lenses in contact: $P =$ _____.

a) $P_1 + P_2$

b) $P_1 - P_2$

c) $1/P_1 + 1/P_2$

d) product

Correct answer: a) $P_1 + P_2$

74. Dispersion: prism deviates _____ most.

a) violet

b) red

c) green

d) yellow

Correct answer: a) violet

75. Scattering explains blue sky: shorter wavelengths _____.

a) scattered more

b) absorbed

c) reflected

d) transmitted

Correct answer: a) scattered more

76. Danger signs red because _____.

a) least scattered

b) most scattered

c) absorbed

d) reflected

Correct answer: a) least scattered

77. Persistence of vision _____.

a) 1/16th second

b) 1 second

c) 1 minute

d) infinite

Correct answer: a) 1/16th second

78. Accommodation power of eye _____.

a) adjust focus

b) fixed

c) colour

d) intensity

Correct answer: a) adjust focus

79. Far-sightedness _____.

a) hypermetropia

b) myopia

c) presbyopia

d) astigmatism

Correct answer: a) hypermetropia

80. Night blindness lacks _____.

a) vitamin A

b) vitamin C

c) protein

d) calcium

Correct answer: a) vitamin A

81. Focal length of eye lens _____.

a) varies 2.5 cm to 4 cm

b) fixed

c) 25 cm

d) infinity

Correct answer: a) varies 2.5 cm to 4 cm

82. Magnifying power of simple microscope _____.

a) D/f

b) f/D

c) $1 + D/f$

d) all

Correct answer: c) $1 + D/f$

83. Telescope angular magnification $m =$ _____.

a) f_o/f_e

b) f_e/f_o

c) D/f

d) $n-1$

Correct answer: a) f_o/f_e

84. Normal adjustment telescope: final image _____.

a) infinity

b) 25 cm

c) retina

d) lens

Correct answer: a) infinity

85. Reflecting telescope uses _____.

a) concave mirror

b) convex lens

c) prism

d) plane mirror

Correct answer: a) concave mirror

86. Plane mirror image distance equals _____.

a) object distance

b) double

c) half

d) zero

Correct answer: a) object distance

87. Boy 3m from plane mirror: distance to image _____.

a) 3m

b) 6m

c) 1.5m

d) 9m

Correct answer: b) 6m

88. Incident angle 30° to mirror surface: angle of incidence _____.

a) 60°

b) 30°

c) 90°

d) 45°

Correct answer: a) 60°

89. Real image can be _____.

a) obtained on screen

b) virtual

c) cannot project

d) erect

Correct answer: a) obtained on screen

90. Object at focus of concave mirror: image _____.

a) infinity

b) at C

c) behind

d) pole

Correct answer: a) infinity

91. Magnification -2 means _____.

a) real, inverted, twice size

b) virtual, erect

c) diminished

d) same size

Correct answer: a) real, inverted, twice size

92. Speed of light in vacuum _____.

a) 3×10^8 m/s

b) 3×10^6 m/s

c) 3×10^{10} m/s

d) 3×10^4 m/s

Correct answer: a) 3×10^8 m/s

93. Glass $n=1.5$: speed of light _____.

a) 2×10^8 m/s

b) 3×10^8 m/s

c) 4.5×10^8 m/s

d) 1.5×10^8 m/s

Correct answer: a) 2×10^8 m/s

94. Critical angle diamond-air _____.

a) 24.4°

b) 41.8°

c) 48.6°

d) 90°

Correct answer: a) 24.4°

95. Power +4 D and -2 D in contact: equivalent _____.

a) +2 D

b) -6 D

c) +6 D

d) 0 D

Correct answer: a) +2 D

96. Microscope maximum magnification when image at _____.

a) near point

b) infinity

c) focus

d) 2F

Correct answer: a) near point

97. Rainbow primary bow: angle _____.

a) 42°

b) 50°

c) 40°

d) 53°

Correct answer: a) 42°

98. Secondary rainbow angle _____.

a) 50°

b) 42°

c) 40°

d) 53°

Correct answer: a) 50°

99. Eye defect corrected by surgery: _____.

a) LASIK

b) glasses only

c) lens replacement

d) all

Correct answer: a) LASIK

100. Optical fibres used in _____.

a) endoscopy, communication

b) telescopes

c) microscopes

d) cameras

Correct answer: a) endoscopy, communication